

Figure 3: Block editor Image 1

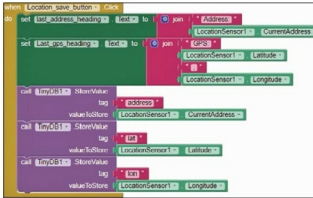


Figure 4: Block editor Image 2

same way shown in the above block editor images. I will explain what each does and how it is called.

1. The blocks in Figures 3, 4 and 5 are easy to understand. I will explain each of them here.
2. The very first block is meant for the location sensor and to keep track of any location changes.
3. As soon as it senses any change in the location, it will display the same against the labels we have put for the address, latitude and longitude.
4. When we click the Save location button, as expected, it should store all available entries in the database.
5. We should save addresses, latitudes and longitudes with their respective tag names for better retrieval when needed.
6. This will update the labels for the last saved location as well.
7. On clicking the Show directions button, our activity starter should trigger Google Maps, and the route between the two GPS points will be shown.
8. We will accomplish that by setting the data URI of the starter activity.
9. Finally, starting the activity will start Google Maps, from out of our application.

Now you are done with the block editor too. Next, we will move to download and install the app on your phone to check how it works.

## Packaging and testing

To test the app, you need to get it on your phone. First, you have to download it to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the **Build** button. It will show you



Figure 5: Block editor Image 3

the option to download the apk to your computer.

2. While downloading, you will be shown the progress and, after it's done, the application will be placed in the download folder of your directory or the location you have set for it.
3. Now, you need to get this apk file to your mobile phone either via Bluetooth or USB cable. Once you have placed the apk file on your SD card, you need to install it. Follow the on-screen instructions to install it. You might get some notification or warning saying: "Install from untrusted source." Allow this from the settings, and after successful installation you will see the icon of your application in the menu of your mobile. Here, you will see the default icon, which can be changed and we will tell you how to do this as we move ahead in this course.

I hope your application is working exactly as per your requirements. Now, depending upon your usability and customisation, you can change various things like the image, sound and behaviour.

## Debugging the application

We have just created the prototype of an app with very basic functionality, but what else might a user be interested in? Let's look at various cases which require serious attention so as not to annoy the user.

1. Can we code it to store more than one location, so that it doesn't overwrite the previous one?
2. Can we share our location with others also, so that they too get to know about our whereabouts?

These are some of the scenarios that might occur, and users will be pretty happy if your app could address these needs too. So think about how you can integrate these into the app. Do ask me if you fail to accomplish any of the above cases. 

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